



A silver lining in MRSA treatment: The synergistic action of poloxamer-stabilized silver nanoparticles and methicillin against antimicrobial resistance

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ABSTRACT

Background: Increasing antibiotic resistance in bacterial infections, including drug-resistant strains like methicillin-resistant *Staphylococcus aureus* (MRSA), necessitates innovative therapeutic solutions. Silver nanoparticles are promising for combating infections, but toxicity concerns emphasize the importance of factors like dosage, size, shape, and surface chemistry. Hence, exploring poloxamer as a stabilizing agent to reduce its toxicity and enhance the antibacterial effect on MRSA is investigated.

Methods: Silver nanoparticles stabilized with poloxamer (AgNPs@Pol) were synthesized through the chemical reduction method and characterized using UV-visible spectrophotometer, HR-TEM, DLS, and Zeta potential measurements. Subsequently, the antibacterial activity of AgNPs@Pol alone and in combination with methicillin against MRSA and methicillin-susceptible *S. aureus* (MSSA) was evaluated using the broth microdilution method. **Results:** AgNPs@Pol showed significant efficacy against MRSA and MSSA, achieving a 100 % reduction in colony-forming units (CFU) at 9.7 µg/ml. The minimum inhibitory concentration (MIC) against MRSA and MSSA was 8.6 µg/ml and 4.3 µg/ml, respectively. A synergistic effect was observed when AgNPs@Pol was combined with methicillin. Treatment with AgNPs@Pol increased reactive oxygen species (ROS) production in both strains, contributing to its antibacterial activity. Real-time qPCR analysis indicated the downregulation of genes involved in antimicrobial resistance and cell adhesion in both strains. Further, the 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) assay demonstrated low cytotoxicity for AgNPs@Pol against MCF-7, MG-63, and NIH-3T3 cell lines.

Conclusion: The developed AgNPs@Pol demonstrated extensive colloidal stability, potent antibacterial activity and synergistic effect with methicillin against MRSA and MSSA. Further studies in primary cells and *in vivo* models may validate its potential for clinical applications.

1. Introduction

The growing concern over antibiotic resistance in bacterial infections, particularly among drug-resistant strains has prompted the search for innovative therapeutic solutions. Notably, methicillin-resistant strains of *Staphylococcus aureus* (MRSA) have emerged as a serious concern in research because they are responsible for various infections ranging from superficial skin and soft tissue infections to severe conditions like endocarditis and osteomyelitis [1–3]. The use of conventional antibiotics requires a higher therapeutic dosage over a longer period, resulting in increased toxicity and a decrease in life

expectancy [4]. Consequently, developing an effective antibacterial agent is a critical challenge in clinically managing MRSA infections. In this light, the researchers have turned their attention to nanoparticle-based disinfection systems, notably silver nanoparticles (AgNPs), for their remarkable antimicrobial properties [5].

AgNPs have emerged as a promising approach in combating bacterial infections with high bactericidal activity. The ability to synthesize AgNPs using various methods to achieve different sizes and shapes has opened new possibilities in medical, consumer product, and water treatment applications [6–9]. The unique antibacterial mechanism of AgNPs involves disrupting bacterial cell membranes, leading to effective

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eradication of harmful pathogens [10–12]. They are widely studied for their antiviral activities and, according to a recent report, green-synthesized AgNPs have shown significant pain-relieving effects (Analgesic activity) in animal models, as well as anticoagulant potential by significantly inhibiting blood clotting [13–15]. AgNPs can be combine with different antibiotics to address multidrug-resistant bacteria by leveraging synergistic, additive, or antagonistic effects [16–20]. Moreover, functionalized AgNPs have been used for drug delivery. For example, a recent study reported that AgNPs functionalized with trans-activator of transcription (TAT) peptides enhance apoptosis in breast cancer cells, demonstrating potential as effective nanocarriers for targeted cancer therapy [21].

However, the debate surrounding the toxicity of AgNPs emphasises the importance of considering various characteristic parameters. The size, shape, dissolution, agglomeration state, chemical composition, specific surface area, crystal structure, surface morphology, surface energy, surface coating, and surface charge, impact biological interactions, fate, and potential outcomes—both desired and adverse—of nanoparticles [22]. Studies have highlighted potential adverse effects of AgNPs on various physiological systems, including the cardiovascular, respiratory, central nervous, and digestive systems [23,24]. The choice of surface stabilizing or capping agents plays a crucial role in determining colloidal stability, diffusivity, and the rate of silver ion release, all of which can influence both the antibacterial efficacy and toxicity of AgNPs. Therefore, selecting the appropriate stabilizing/capping agent for the surface functionalization of nanoparticles is paramount in enhancing the unique characteristics that are useful in biomedical applications, while minimizing potential harm [21,25].

Poloxamers are inert amphiphilic tri-block copolymers known for their low toxicity and biodegradability. These compounds have recently demonstrated promise in wound healing as an antibacterial agent. It has been tested for mildness and less irritation upon topical application and can be used orally and parenterally [26,27]. Moreover, Poloxamer surfactants have received approval for pharmaceutical use, further supporting their viability for clinical applications. Hashemi et al. recently reported the incorporation of ceragenin (CSA-131), a synthetic antimicrobial peptide used to treat chronic lung infections, into poloxamer micelles to protect lung cilia while maintaining antimicrobial efficacy [28]. Poloxamer has also been extensively researched for its potential applications in drug delivery and mucosal treatments [29,30]. Notably, they can reverse multidrug resistance (MDR) in cancer cells by inhibiting drug efflux transporters responsible for expelling drugs from the cells. This unique property enhances the effectiveness of chemotherapy drugs by increasing their accumulation within the cells and intensifying their toxicity. Consequently, poloxamer hold promise as a drug delivery platform, capable of overcoming MDR and enhancing the treatment outcomes [31].

In this context, the synthesis of AgNPs@Pol stabilized with poloxamer as a potential antibacterial agent against MRSA strains is investigated, aiming to address the increasing antibiotic resistance of MRSA and improve the antimicrobial activity while minimizing cytotoxicity.

2. Materials and methods

2.1. Bacterial isolates, cell lines, chemicals, reagents, antibacterial agents, and media

Sodium borohydride (NaBH₄, Extrapure AR, ACS), sodium chloride (NaCl, Extrapure AR, 99.9 %), sodium phosphate monobasic (Extrapure AR, 99 %), and sodium phosphate dibasic (Extrapure AR, 99 %) were purchased from SRL (Maharashtra, India). Poloxamer 407 (Purified, non-ionic), methicillin sodium salt, resazurin sodium salt, 2',7'-dichlorofluorescein diacetate (DCFDA), 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT), 1-Hydroxybenzotriazole hydrate (Trizol), and dimethyl sulfoxide (DMSO) were purchased from Sigma Aldrich (Karnataka, India). Tris-base, Triton X-100, EDTA

disodium salt dihydrate, Luria Bertani (LB) broth, Mueller-Hinton (MH) broth, mannitol salt agar (MSA), Baird Parker agar (BPA), blood agar, MeReSa agar medium, and antibiotic disc (Amikacin, linezolid, methicillin, norfloxacin, oxacillin, penicillin – G, streptomycin, and tetracycline) were purchased from Himedia (Maharashtra, India). Silver nitrate (AgNO₃, Crystalline/Certified ACS), Alpha-modified Eagle's medium (α-MEM), Dulbecco's Modified Eagle Medium (DMEM), Fetal bovine serum (FBS), sodium pyruvate, non-essential amino acids, antibiotic-antimycotic solution, and gentamycin sulfate were purchased from Gibco (Thermo Fisher Scientific, Maharashtra, India).

The MRSA strain (ATCC BAA-44) was purchased from Himedia (Maharashtra, India) and the Methicillin-sensitive *Staphylococcus aureus* (MSSA) was obtained from clinical repository of K. S Hegde Hospital, Nitte (Deemed to be University), Mangalore, India. The cell lines used in the study, human breast adenocarcinoma cells (MCF-7), osteosarcoma cells (MG-63), and mouse embryonic fibroblasts (NIH-3T3) were procured from National Centre for Cell Sciences (NCCS) Pune (Maharashtra, India).

2.2. Synthesis and characterization of AgNPs@Pol

The poloxamer stabilized AgNPs@Pol was prepared by chemical reduction of silver nitrate using NaBH₄ as a reducing agent. Briefly, a volume of 2.5 ml of 10 mM AgNO₃ added drop-wise to 47 ml of Millipore water containing NaBH₄ (Final concentration of 1 mM in 50 ml) under vigorous stirring in an ice bath on a digital hotplate with magnetic stirrer (900.DNDG.01, Glassco, Maharashtra, India) followed by stabilization with the addition of 0.5 ml of 100 mM poloxamer (AgNO₃: NaBH₄: Poloxamer = 0.5:1:1). The as-synthesized AgNPs@Pol was centrifuged at 10000 rpm at 4 °C for 20 min using refrigerated centrifuge (Eppendorf 5810R, Eppendorf AG, Hamburg, Germany) and re-dispersed in 1 mM poloxamer solution, and stored at 4 °C for further studies.

The UV-visible spectrophotometer (Multiskan sky, Thermo Fisher Scientific, Singapore) was used to measure the plasmonic absorbance of the AgNPs@Pol. The size of the AgNPs@Pol was determined using high-resolution transmission electron microscopy (HR-TEM, LaB6 Electron Gun-Transmission Electron Microscope 200 Kv (JEM-2100) from JEOL Ltd. (Tokyo, Japan)) at STIC India, Kochi and dynamic light scattering spectroscopy (DLS, Horiba SZ-100V2 (Kawasaki, Japan)) at NUCSER, Nitte University. Zeta potential measurements were used to determine the surface charge of the AgNPs@Pol (Horiba SZ-100V2, Horiba Ltd. (Kawasaki, Japan)) at NUCSER, Nitte University. Silver ion (Ag⁺) concentration in AgNPs@Pol were determined using inductively coupled plasma-optical emission spectroscopy (ICP-OES, PerkinElmer Optima 5300 DV, PerkinElmer (Massachusetts, USA)) at IIT Madras. A calibration curve of elemental Ag concentration vs AgNPs@Pol absorbance was plotted and used as the standard calibration curve for AgNPs@Pol concentration in subsequent studies.

2.3. Antibacterial activity, time-kill curves, and resazurin assay

The identification and characterization of the MRSA and MSSA bacterial isolates were performed (methods and results described in supplementary information). We conducted all antibacterial studies using MH, widely used for bacterial susceptibility testing by clinical standards [32]. LB broth/agar was utilized for general bacterial culture and colony-forming unit (CFU) determination. The spread-plate count method was used to determine the time-kill curves and minimum inhibitory concentration (MIC) of AgNPs@Pol against the bacterial isolates. Briefly, 500 µl of MH was taken in 1.5 ml of microcentrifuge tubes, and 10⁵ CFU/ml of the bacterial cultures were inoculated separately. The AgNPs@Pol was added in five different concentrations (Final concentration of 1.2, 2.4, 4.8, 7.3, and 9.7 µg/ml) and made up to 1 ml by adding autoclaved double distilled water. Methicillin was used as the antibiotic control (Final concentration of 1, 3, 5, and 10 µg/ml dissolved in 1 × PBS buffer with pH 7.4). Bacterial cultures without AgNPs@Pol

were used as the negative control. The samples were prepared in four sets for four different time intervals (0, 4, 6, and 8 hours) in triplicates and incubated in a static incubator at 37 °C (Equitron general incubator, Equitron Medica Pvt. Ltd., Maharashtra, India). Further, the cultures were serially diluted, spread-plated onto the LB agar and incubated for 24 hours at 37 °C. The CFU was obtained using equation (S1).

The MIC was determined using a resazurin-based microtiter plate assay [33]. Briefly, the microtiter plate wells were added with 100 µl of MH broth. Serial dilutions of AgNPs@Pol were achieved by adding 100 µl of AgNPs@Pol to each well, resulting in the first and the last column of the dilution with highest and lowest concentration of AgNPs@Pol, respectively. Following that, 100 µl of MH broth containing the bacterial culture (MRSA or MSSA) with an optical density of 0.005 at OD_{600 nm} (~10⁶ CFU/ml) was added to each well and incubated at 37 °C in a static incubator. Bacterial cultures without AgNPs@Pol served as the control, while MH broth alone was used as the negative control. After 8 hours, 20 µl of a 0.025 % solution of resazurin dye in water was added to each well and incubated at 37 °C in dark conditions. The incubation continued until the bacterial controls displayed a complete color change from purple to pink. The absorption was measured at 570 nm using a UV-visible spectrophotometer (Multiskan sky, Thermo Fisher Scientific, Singapore). The MIC of AgNPs@Pol was determined as the lowest concentration of the test at which the color change occurred. The same protocol was used to determine the MIC of methicillin.

2.4. Combination assay and fractional inhibitory concentration (FIC) index calculations

The combination assays for the AgNPs@Pol and methicillin, a checkerboard method was followed with slight modifications [34,35]. The concentration of AgNPs@Pol (1.2, 2.4, 4.8, 7.3, and 9.7 µg/ml) and methicillin (1, 3, 5, and 10 µg/ml) was varied in all plausible combinations, and the CFU was determined after 8 hours of incubation using the procedure described in section 2.3 above. The FIC index of the combination assay was obtained using the following equation (1):

$$FIC\ index = FIC_X + FIC_Y = \frac{[X]}{[MIC]_X} + \frac{[Y]}{[MIC]_Y} \quad \text{Eq. (1)}$$

Where X/Y is the drug concentration in the given sample and MIC_X/MIC_Y is the control MIC of the organism to the drug alone. The FIC index for a sample was calculated as the sum of the FICs for both AgNPs@Pol and methicillin present in the sample. And an FIC mean was derived for each checkerboard by averaging the FIC index values of all the samples.

2.5. Estimation of reactive oxygen species using DCFDA assay

The reactive oxygen species (ROS) of AgNPs@Pol and methicillin-treated MRSA and MSSA samples were determined using a modified DCFDA assay [36]. Briefly, 10⁵ CFU/mL of MRSA and MSSA isolates were treated with sub-MIC concentrations of AgNPs@Pol and methicillin in combination and alone (Final volume of 1 ml in 1.5 ml centrifuge tubes) in triplicates and incubated at 37 °C for 5 hours in a static incubator. Subsequently, the samples were washed thrice with 1x PBS by centrifuging at 10,000 rpm for 2 min at 4 °C. The obtained bacterial pellets were re-dispersed in 350 µl of 1x PBS and sonicated using a probe sonicator (6 mm, Qsonica Q500 Sonicator, Qsonica, LLC, New York, USA) for 20 seconds at 20 % amplitude, repeated three times. This was followed by adding 1 µl of 10 µM DCFDA and incubating at room temperature for 30 min. After the incubation, the supernatant of the samples was collected by centrifugation at 10,000 rpm for 2 min at 4 °C. The ROS levels were quantified by assessing fluorescence emission intensity of the sample in a quartz cuvette at 524 nm by exciting at 498 nm using a spectrofluorometer (JASCO FP-8300, JASCO Corporation, Tokyo, Japan; Slit width: 2.5–5 nm; Sensitivity: High; Scan speed: 500 nm/min) [37].

ROS estimation is also affected by the number of bacterial cells present in the sample. Therefore, we compared the intracellular ROS production of treated samples with the untreated control after normalization for cell viability. The cell viability of the treated and untreated samples was determined using the resazurin assay (described in section 2.3) with slight modification. In short, resazurin-treated samples were diluted in 1 ml of 1x PBS in 1.5 ml centrifuge tubes, and their fluorescence emission was measured by exciting at 570 nm in a quartz cuvette using JASCO spectrofluorometer (Slit width: 2.5–5 nm; Sensitivity: Low; Scan speed: 500 nm/min). The fluorescence intensity obtained from resazurin assay for bacteria without the treatment were used as the positive control (representing 100 % cell viability), while media without bacteria served as the negative control (representing 0 % cell viability). Thus, the fluorescence intensity of the DCFDA assay was normalized to the cell viability using the following equation (2):

$$\text{Normalized ROS} = \frac{R}{C} \times 100 \quad \text{Eq. (2)}$$

Where, R is the fluorescence intensity obtained from DCFDA assay, and C is the cell viability (%) obtained using resazurin assay. The relative fluorescence intensity of the DCFDA was then determined using equation (3):

$$\text{Relative ROS} = \frac{\text{Normalized ROS for treated}}{\text{Normalized ROS for untreated}} \quad \text{Eq. (3)}$$

2.6. Gene expression analysis by real-time polymerase chain reaction (qPCR)

Freshly revived bacterial strains, i.e. MRSA and MSSA were cultured for overnight. The overnight cultures, adjusted to a bacterial load of 10⁵ CFU/ml, were exposed to sub-MIC concentrations of AgNPs@Pol, methicillin, and a combination of both. Bacterial culture without treatment was used as a control. After 6 hours of incubation at 37 °C in a static incubator, the pellets were harvested by centrifugation at 13000 rpm at 4 °C for 2 min. The pellets were flash frozen in liquid nitrogen and stored at - 80 °C until further processing. The frozen cell pellet was resuspended in 100 µl of Tris-EDTA buffer containing 0.1 % of TritonX-100. To this lysozyme is added (50 µg/ml) and incubated at 37 °C for 30 min with intermittent mixing for the improved lysis of the hard-to-lyse bacterial strains. Subsequently, the standard total RNA extraction was performed by Trizol method [38]. The isolated RNA pellets were air-dried and resuspended in RNase-free water and stored at - 80 °C.

The regulation in expression of genes related to antibiotic resistance and biofilm formation after treating with AgNPs@Pol, methicillin, and in combination was analyzed using One Step Prescript™ RT-PCR Kit (Perfect Real Time, RR064A, Takara Bio Inc, Shiga, Japan) and Bio-Rad CFX Opus 96 qPCR instrument (Bio-Rad Laboratories, California, United States). The primer sequences used for qPCR were shown in Table S6. A volume of 10 µL reaction was set for each sample as given in the kit. The templates were amplified with an initial denaturation of 95 °C for 2 min, followed by 40 cycles of 95 °C for 15 seconds, 55 °C for 60 seconds, and 72 °C for 30 seconds. The relative gene expression (fold change) was calculated using the method 2^{-ΔΔCT}, normalized against the *rpoB* as a housekeeping gene and compared with the untreated control group. The data was analyzed using CFX Maestro Software and Microsoft Excel 2019.

2.7. Cytotoxicity assessment using MTT assay

1-(4,5-Dimethylthiazol-2-yl)-3,5-diphenyl formazan (MTT) assay was used to assess the cytotoxicity of AgNPs@Pol on cells. MCF-7 and NIH-3T3 were maintained in DMEM-High Glucose and MG-63 in α-MEM with sodium pyruvate and non-essential amino acids (1 mM each). Both media were supplemented with 10 % fetal bovine serum, 1 % antibiotic-antimycotic solution, and gentamycin sulfate. The cells were incubated

at 37 °C with 5 % CO₂ under humid conditions (IncuSafe CO₂ Incubators, PHCbi, Nagoya, Japan).

Following that, cells were grown in a 96-well plate at a density of 7×10^3 cells/well. AgNPs@Pol of concentrations ranging from 1 to 20 µg/ml were treated in the cells. After 24 hours, the media was replaced with 100 µl media containing MTT reagent (0.5 mg/ml) and incubated for 4 hours at 37 °C. Subsequently, the media was removed, and 100 µl of DMSO was added to each well to dissolve the formazan crystals formed [39]. The absorbance of the plate was read at 570 nm using a UV–visible spectrophotometer (Multiskan sky, ThermoScientific, Singapore). Cell viability was calculated by normalizing treated cells to untreated cells (representing 100 % cell viability).

2.8. Statistical analysis

All the experiments were performed in triplicate unless otherwise specified. The statistical significance of variance (one-way ANOVA) was assessed using GraphPad Prism 9.0 software. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$.

3. Results

3.1. Poloxamer ensures uniform particle size distribution and superior aqueous stability of AgNPs@Pol

AgNPs@Pol was prepared by simple chemical reduction using NaBH₄ followed by stabilization with non-ionic poloxamer. The optical properties, size, and surface charge of the AgNPs@Pol were characterized using the UV–visible spectrophotometer, DLS, HR-TEM, and Zeta potential measurements. The UV–visible spectrophotometer analysis revealed distinct plasmonic peaks at 391 nm [40] (Fig. 1a) and the hydrodynamic diameter obtained through DLS, indicated size of 11.5 ± 0.12 nm for AgNPs@Pol (Fig. 1b). The size of the AgNPs@Pol obtained from HR-TEM analysis was 10.5 ± 2 nm (Fig. 1c), which is in good agreement with the DLS analysis.

Furthermore, Zeta potential measurements indicated surface charges of 2.5 mV at pH 4.0, - 0.8 mV at pH 8.0, and - 21.3 mV at pH 10.0, suggests the effective surface stabilization of the AgNPs@Pol. The absorption spectrum was monitored for 3 months to assess the stability of AgNPs@Pol over time (Fig. S1). It was noted that the AgNPs@Pol showed excellent stability, as evidenced by no significant change in

plasmonic absorbance wavelength and intensity. However, a broadening of the plasmonic peak at 454 nm was observed after 3 months, potentially attributed to the change in particle size, as reported previously [40].

3.2. Time-kill assay and resazurin assay showed the antibacterial effect of AgNPs@Pol and methicillin against MRSA and MSSA

The time-dependent growth of MRSA without and with AgNPs@Pol/methicillin treatment was monitored at different time intervals (0, 4, 6, and 8 hours). The results are presented in Fig. 2 as colony counts. After 3 hours of incubation, no MRSA bacterial cell growth was observed at all concentrations except at 1.2 µg/ml of AgNPs@Pol treatment, which showed a 66.5 % reduction in CFU compared to the untreated MRSA control ($p < 0.0001$) (Fig. 2a). Compared to the untreated MRSA control, AgNPs@Pol demonstrated more than 44.97 % reduction in CFU after 6 hours and more than 39.38 % reduction in CFU after 8 hours of induction. At a concentration of 9.7 µg/ml of AgNPs@Pol, a 100 % reduction in CFU was seen across all time intervals when compared to the untreated MRSA control ($p < 0.0001$). This finding suggests that AgNPs@Pol alone is effective at killing multidrug-resistant MRSA bacteria.

According to the CLSI guidelines, MRSA is characterized by an oxacillin/methicillin MIC ≥ 4 µg/ml [41]. As a result, the concentration range of methicillin used in our study ranged from 1 to 10 µg/ml. Methicillin decreased the CFU after 3 hours of treatment (Fig. 2b). The percentage reduction in CFU for methicillin concentrations of 1, 3, 5, and 10 µg/ml were 33.53 %, 77.63 %, 94.05 %, and 100 %, respectively, compared to the untreated MRSA control ($p < 0.0001$). However, after 8 hours, the percentage reduction of CFU decreased to 32.56 %, 48.29 %, 57.88 %, and 68.99 % for methicillin concentrations of 1, 3, 5, and 10 µg/ml, respectively.

Fig. 3a illustrates the results obtained from exposure to both MRSA and MSSA with different concentrations of AgNPs@Pol for 8 hours. At lower concentrations of AgNPs@Pol (1.2, 2.4, and 4.9 µg/ml), the percentage reduction in CFU for MSSA (51.65 %, 77.94 %, and 82.88 %, respectively) was more pronounced than for MRSA (39.38 %, 58.23 %, and 68.02 %, respectively). At a higher concentration of AgNPs@Pol (9.7 µg/ml), a 100 % reduction was observed for both MRSA and MSSA compared to untreated bacterial control ($p < 0.0001$).

In Fig. 3b, the percentage reduction in CFU for MRSA was observed

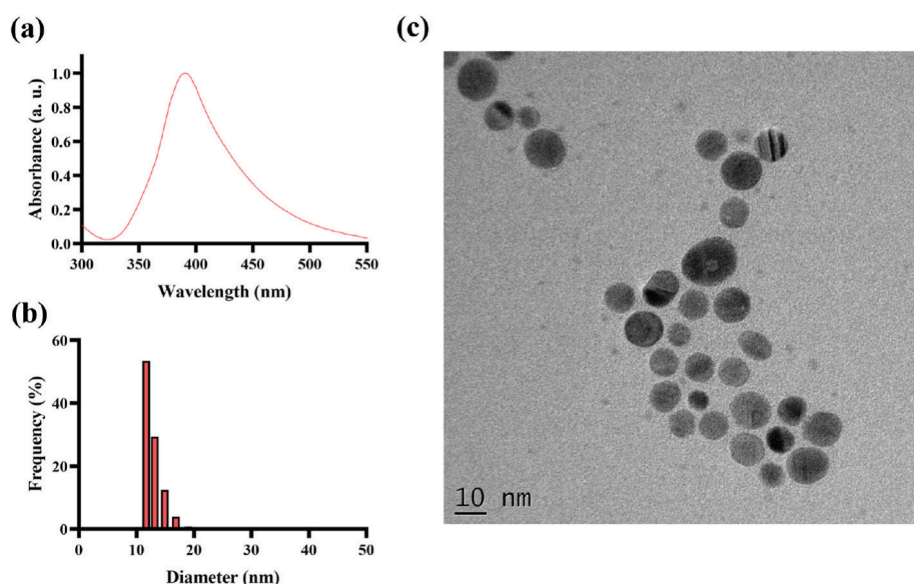


Fig. 1. (a) The plasmonic absorption spectrum obtained for the AgNPs@Pol using UV–visible spectrophotometer (391 nm); (b) the hydrodynamic diameter obtained from DLS analysis (11.5 ± 0.12 nm); and (c) the HR-TEM image obtained for AgNPs@Pol (10.5 ± 2 nm).

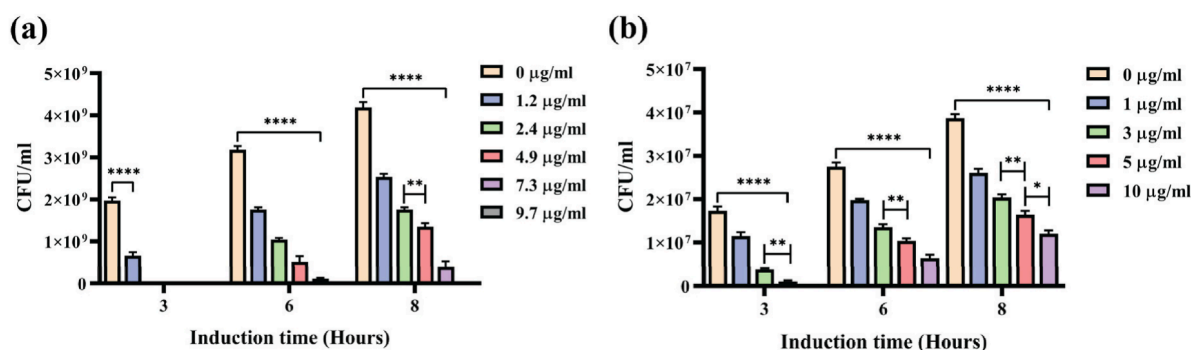


Fig. 2. The antibacterial effect of (a) AgNPs@Pol (0–9.7 µg/ml) and (b) methicillin (0–10 µg/ml) against MRSA was investigated at different time intervals (0, 3, 6, and 8 hours) plotted as mean $\pm$ SD, $n = 3$. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$ (one-way ANOVA).

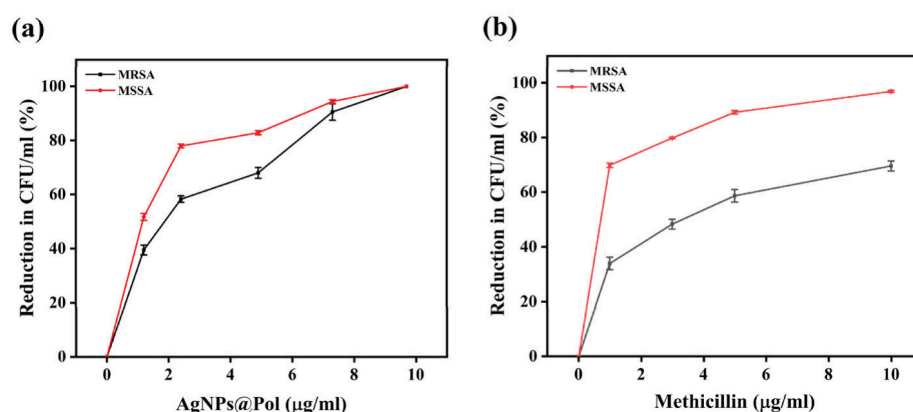


Fig. 3. The comparison of antibacterial effect of different concentrations of (a) AgNPs@Pol (0–9.7 µg/ml) and (b) methicillin (0–10 µg/ml) against MRSA and MSSA after inducing for 8 hours plotted as mean $\pm$ SD, $n = 3$. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$ (one-way ANOVA).

to be 32.55 %, 47.29 %, 57.88 %, and 68.99 % when treated with methicillin concentrations of 1, 3, 5, and 10 µg/ml, respectively, compared to the untreated bacterial control ($p < 0.0001$). Similarly, the percentage reduction in CFU for MSSA was found to be 69.72 %, 79.82 %, 89.24 %, and 96.87 % for the corresponding concentrations of methicillin.

The resazurin assay was used to determine the MIC of AgNPs@Pol and methicillin against MRSA and MSSA isolates. AgNPs@Pol showed an MIC of 8.6 µg/ml for MRSA, which was higher than the MIC of 4.3 µg/ml for MSSA (Fig. 4a). The MIC of methicillin for MSSA was 10 µg/ml

(Fig. 4b). In contrast, MRSA showed complete resistance against the maximum concentration of methicillin used.

3.3. The combination assay demonstrated a synergistic effect of AgNPs@Pol and methicillin against MRSA and MSSA

A modified checkboard method was adopted to determine the combined effect of AgNPs@Pol in combination with methicillin against MRSA and MSSA isolates [34]. The combination assay was performed, maintaining the concentration of methicillin at 1 µg/ml, while

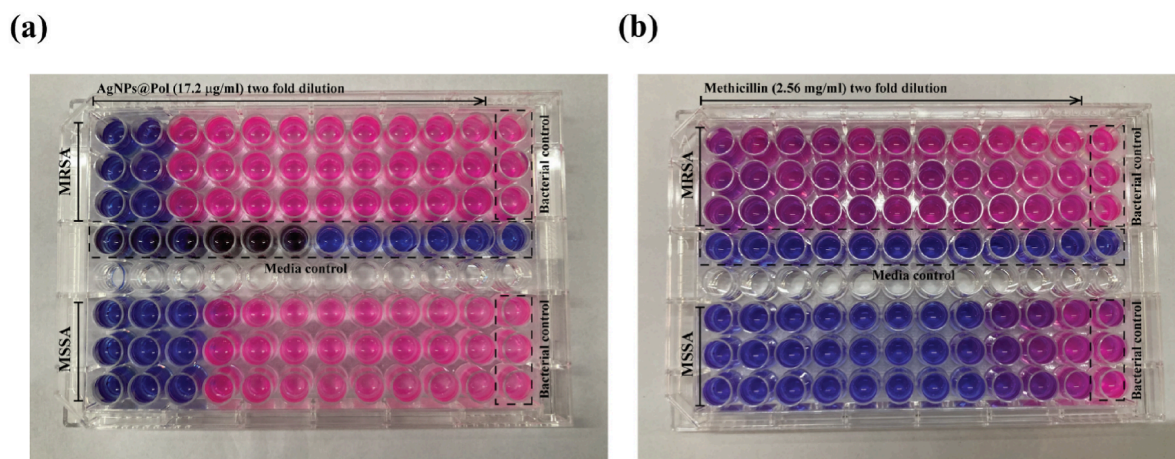


Fig. 4. The resazurin staining of MRSA and MSSA isolates in the absence and presence of (a) AgNPs@Pol and (b) methicillin. The MIC of both AgNPs@Pol and methicillin were determined as the lowest concentration of the test at which the colour change occurred from purple to pink.

concentrations of AgNPs@Pol were varied (0–9.7 $\mu\text{g/ml}$). The results showed that the combination led to a substantial reduction in the CFU of MRSA, as shown in Table S4. Notably, the decrease in CFU was more pronounced in MSSA compared to MRSA. This could be due to the inherent sensitivity of MSSA to methicillin (Fig. 3b and 4b). In comparison to the untreated bacterial control, when the concentration of AgNPs@Pol exceeded 7.3 $\mu\text{g/ml}$, the percentage reduction in CFU of MRSA was more than 90 %, regardless of whether it was used in combination with methicillin or not ($p < 0.0001$).

The fractional inhibitory concentration (FIC) index is the most common method used to measure drug interaction in a combination assay. Because multiple definitions and ways for interpreting the FIC index are provided in the literature. The mean FIC index was calculated for each checkboard by averaging the FIC index values obtained for each sample with varied drug combinations (Fig. 5).

The interaction was classified as synergistic if the FIC index was <1 , additive if it was equal to 1, and antagonistic if it was >1 [35]. According to equation (1), the FIC index for AgNPs@Pol in combination with methicillin was 0.9, indicating a synergistic mode of action (~ 1).

3.4. Synergistic effect of AgNPs@Pol and methicillin increased the intracellular ROS levels in both MRSA and MSSA strains

DCFDA assay measured intracellular ROS production following treatment with AgNPs@Pol alone and combined with methicillin. Fig. 6a shows no significant difference in DCFDA fluorescence between the untreated and methicillin-treated MRSA samples since it is resistant to methicillin. In contrast, methicillin treatment increased ROS levels significantly in MSSA ($p < 0.0001$) due to its antibiotic susceptibility (Fig. 6b). AgNPs@Pol significantly elevated the production of ROS in both MRSA and MSSA compared to the untreated control ($p < 0.0001$). Furthermore, the combination of AgNPs@Pol and methicillin notably increased ROS levels in MRSA ($p = 0.0302$) and MSSA ($p = 0.0142$). Additionally, the impact of AgNPs@Pol treatment alone and in combination with methicillin on ROS production was more pronounced in MSSA than in MRSA.

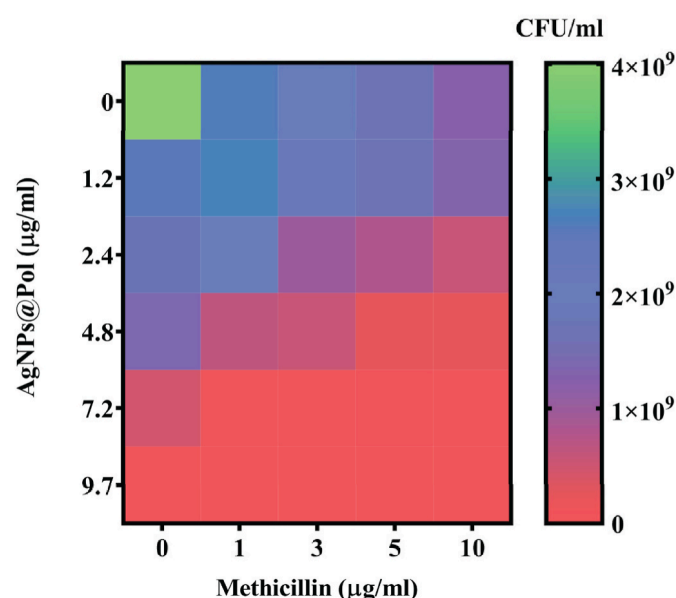


Fig. 5. The combination assay of AgNPs@Pol and methicillin against MRSA using a modified checkerboard method plotted as mean $\pm$ SD, $n = 3$. Green boxes signify higher growth of MRSA (Bacterial control). A dark red hue denotes diminished bacterial growth. Meanwhile, a gradient from blue to violet represents a progressive decrease in bacterial growth.

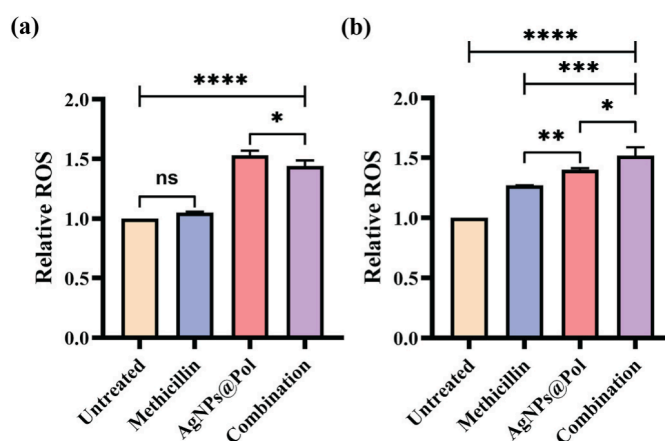


Fig. 6. Relative ROS obtained for (a) MRSA and (b) MSSA after treating with AgNPs@Pol alone and in combination with methicillin plotted as mean $\pm$ SD, $n = 3$. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$ (one-way ANOVA).

3.5. AgNPs@Pol treatment significantly affects the gene expression associated with antimicrobial resistance (AMR)

The effects of AgNPs@Pol, methicillin, and combination on the gene expression of MRSA and MSSA were investigated using real-time qPCR. Following treatment with methicillin, MRSA showed a substantial upregulation in the expression of *mecA* ($p < 0.0001$), *mecR1* ($p = 0.0083$), and *femA* ($p = 0.0061$) genes compared to the untreated control (Fig. 7a, b, and 7c). MSSA, being a methicillin-sensitive bacterium, did not exhibit any expression of these genes. Treatment with AgNPs@Pol alone significantly decreased the expression levels of *mecA*, *mecR1*, and *femA* genes in MRSA (*mecA*: $p < 0.0001$; *mecR1*: $p = 0.0135$; *femA*: $p = 0.0356$). Similarly, when AgNPs@Pol was used in combination with methicillin, there was a substantial reduction in the expression of these genes (*mecA*: $p < 0.0001$; *mecR1*: $p = 0.0116$; *femA*: $p = 0.0023$), compared to the untreated MRSA control. This implies that the AgNPs@Pol with or without methicillin significantly downregulates gene expressions associated with AMR.

When MRSA was exposed to AgNPs@Pol, there was a 2.8-fold increase in the expression of the *blaI* gene compared to methicillin treatment alone (Fig. 7d). However, the combination treatment did not affect the expression of the *blaZ* inhibitor. The unaltered expression of *blaI* in the combination treatment could be attributed to the presence of methicillin, which may induce a minimal level of beta-lactamase expression. Interestingly, in MSSA, treatment with AgNPs@Pol resulted in a 2-fold upregulation of *blaI* compared to the untreated control ($p = 0.0071$). Both AgNPs@Pol alone ($p = 0.0010$) and in combination with methicillin ($p = 0.0008$) were found to reduce the expression of the *icaA* gene significantly compared to MRSA treated with methicillin alone (Fig. 7e). In MSSA, the *icaA* gene was completely downregulated in all treatment conditions (Fig. S5), possibly due to the sensitivity of the strain towards methicillin. In addition, the expression of the *sarA* gene was significantly downregulated for the AgNPs@Pol alone and the combination treatment in both MSSA and MRSA (Fig. 7f).

3.6. Cytotoxicity assessment of AgNPs@Pol using MTT assay

Although AgNPs@Pol have demonstrated effectiveness in killing MRSA and MSSA in this study, it is essential to evaluate their cytotoxicity as well. Therefore, the cytotoxicity of AgNPs@Pol on MCF-7 (human breast cancer), MG-63 (human osteosarcoma), and NIH-3T3 (mouse fibroblast) cell lines was evaluated through the MTT assay (Fig. 8). The results indicated that the degree of cytotoxicity exhibited by AgNPs@Pol was dose-dependent. It was observed that the effect of

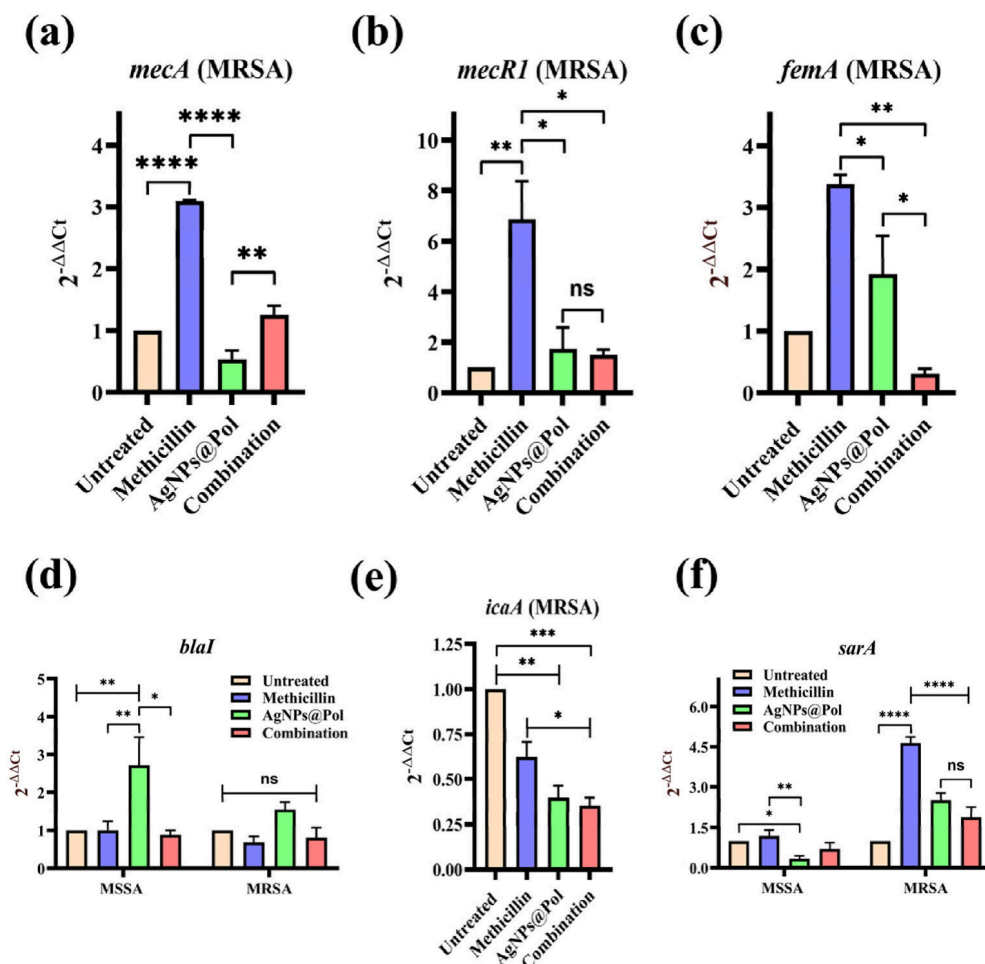


Fig. 7. The gene expression levels in MRSA and MSSA were determined after treating with sub-MIC concentrations of AgNPs@Pol and methicillin alone and in combination using qPCR: (a) *mecA*, (b) *mecR1*, and (c) *femA*, (d) *blaI*, (e) *icaA*, and (f) *sarA*. The data plotted as mean $\pm$ SD, $n = 3$. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$ (one-way ANOVA).

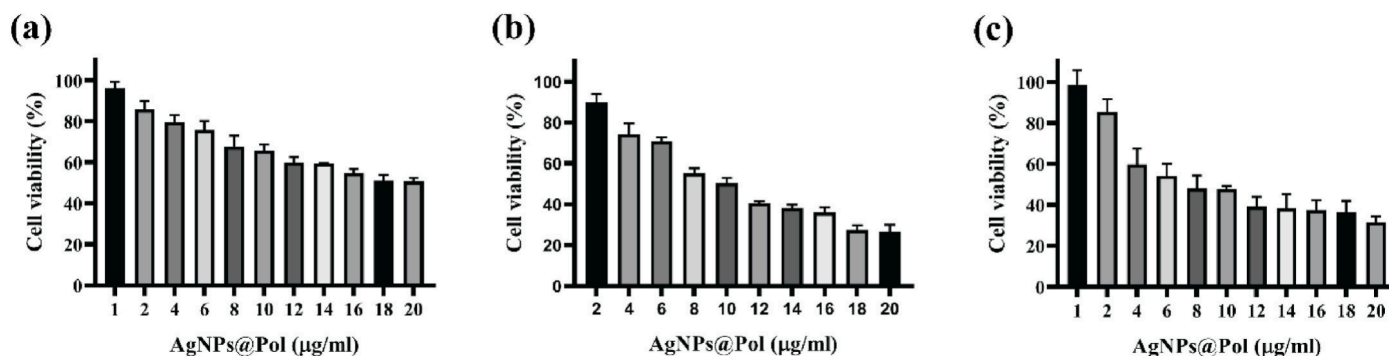


Fig. 8. Cell viability of (a) MCF-7, (b) MG-63, and (c) NIH-3T3 cells treated with varying concentrations of AgNPs@Pol plotted as mean $\pm$ SD, $n = 3$.

AgNPs@Pol was more significant in NIH-3T3 cells compared to MCF-7 and MG-63 cells. It was also noted that concentrations up to 6 $\mu\text{g/ml}$ for MCF-7, 4 $\mu\text{g/ml}$ for MG-63 cells, and 2 $\mu\text{g/ml}$ for NIH-3T3 cells did not show significant cytotoxicity, but with cell viability was $\sim 80\%$.

4. Discussion

The emergence of antibiotic resistance in MRSA has necessitated the investigation of novel treatment approaches [1–4]. AgNPs as an antibacterial nanomaterial have shown promise in combating bacterial infections via their ability to disrupt bacterial cell membranes and induce

antimicrobial activities [5–20]. However, their potential toxicity necessitates careful consideration of factors like dosage, size, shape, and surface chemistry. Using stabilizing agents can influence both the antibacterial efficacy and toxicity of AgNPs [23,24]. Poloxamers, low-toxic and biodegradable tri-block copolymers, have shown potential in wound healing and drug delivery, including reversing MDR in cancer cells [26–31]. Therefore, employing poloxamer as a stabilizing agent may help to reduce toxicity, improve biodegradability, and enhance the antibacterial properties of AgNPs. In this study, we synthesized poloxamer-stabilized AgNPs@Pol to evaluate its efficacy against multi-drug-resistant MRSA. AgNPs@Pol exhibited a uniform size

distribution and remarkable colloidal stability over three months as evidenced by plasmonic absorption spectra and Zeta potential analysis.

The results obtained from our study indicate that AgNPs@Pol have substantial antibacterial effects against both MRSA and MSSA strains. However, the effectiveness of methicillin against MRSA diminished over time, likely due to the antibiotic resistance of the *S. aureus* isolate used in the study [42]. A remarkable difference was observed in the MIC values for MRSA and MSSA when treated with AgNPs@Pol, with MRSA demonstrating a higher MIC. This observation was matching with the previous findings of Saber et al., who reported the antibacterial activities of starch-capped AgNPs against both MRSA and MSSA [43]. However, it is noteworthy that a complete reduction in CFU was observed for both MRSA and MSSA at a concentration of 9.7 µg/ml and (MIC for MSSA) of AgNPs@Pol, respectively, highlighting its potential in eradicating bacterial isolates. Interestingly, a combination of AgNPs@Pol with methicillin markedly reduced the CFU of MRSA, even at a low methicillin concentration of 3 µg/ml. The FIC index of 0.9 suggests a synergistic effect between AgNPs@Pol and methicillin, enhancing the potential of this combination as a novel therapeutic approach for treating MRSA infections [35]. This synergistic effect could help to reduce the risk of side effects and prevent the development of antibiotic resistance.

The primary mechanism of cytotoxicity for AgNPs@Pol against bacterial cells is the generation of ROS [44]. This process induces oxidative stress within cells, damaging proteins and nucleic acids. Our study found a significant increase in ROS levels in both MRSA and MSSA upon treatment with AgNPs@Pol alone. Interestingly, MRSA, known for its methicillin resistance, does not generate considerable levels of ROS upon methicillin treatment. However, a combination of methicillin and AgNPs@Pol led to a significant rise in ROS production, implying a synergistic effect between the two. Previous reports indicate that heavy metals in conjunction with antibiotics can enhance ROS generation [45]. Furthermore, we observed a difference in ROS production between MRSA and MSSA upon treatment with AgNPs@Pol, aligning with the disparity observed in the MIC. These differences in ROS generation may be due to the distinct metabolic states of each strain and the effects of antibiotics on their subcellular processes [46].

Considering the difference between MRSA and MSSA in the context of drug-resistant genes, it is crucial to further investigate how the treatment affects the expression of these genes. The *mecA* gene and its cognate regulatory gene, *mecR1*, are of significant interest in MRSA. The *mecA* gene encodes for a penicillin-binding protein, PBP2a, which has a significantly reduced affinity for β -lactam antibiotics such as methicillin [3]. Conversely, *mecR1* positively regulates the expression of *mecA* [47, 48]. The gene *femA* is involved in precursor synthesis for peptidoglycans, influencing methicillin resistance in MRSA. These genes also serve as molecular markers in the identification of MRSA [49]. The real-time qPCR analysis revealed that both AgNPs@Pol alone and in combination with methicillin effectively reduce the expression of genes *mecA*, *mecR1*, and *femA*. This implies that AgNPs@Pol could potentially inhibit the development and spread of methicillin resistance in *S. aureus*. MSSA, being a methicillin-sensitive bacterium, did not exhibit gene expression for *mecA*, *mecR1*, and *femA* [50].

In *S. aureus*, the *blaZ* gene regulates the production of beta-lactamase, while *blaI* suppresses the expression of *blaZ* [51]. Both MRSA and MSSA showed an increased expression of *blaI* after AgNPs@Pol treatment. The increased expression of *blaI* in MSSA is likely due to the suppression of the *blaZ*-*blaR1*-*blaI* system, thereby reducing the beta-lactamase activity, as previously reported [52]. Bacterial aggregation and attachment are important aspects of *S. aureus* for its survival under stress conditions. The *icaA* gene in *S. aureus* is involved in synthesizing capsular polysaccharides intercellular adhesins (PIA), an amino sugar-rich polysaccharide essential for cell-to-cell adhesion during biofilm formation [53]. Meanwhile, *sarA* positively regulates the transcription of *icaA* [54]. A recent study has shown that treating MRSA with *Myrtus communis* L. leads to the downregulation of the *icaA* and

sarA genes, which in turn inhibits biofilm formation [55]. Consequently, the significant downregulation of the gene *icaA* and *sarA* suggests that AgNPs@Pol effectively disrupt the cell-to-cell adhesion processes essential for biofilm formation.

The cytotoxicity of the AgNPs@Pol was evaluated across three different cell lines using the MTT assay. The results indicate that AgNPs@Pol, at concentrations of up to 6 µg/ml, exhibited low cytotoxicity cell viability when treated in MCF-7 cell lines. This viability was significantly higher than that observed with biologically synthesized AgNPs (3 µg/ml) using a cell-free extract from *Polysiphonia algae*, as reported in a previous study [56]. However, in the case of MG-63 and NIH-3T3 cell lines, AgNPs@Pol demonstrated low cytotoxicity at concentrations of 4 µg/ml and 2 µg/ml, respectively. It was reported that the cytotoxicity of AgNPs can vary across different cell types due to their different physiological characteristics [57], as demonstrated in MCF-7, MG-63, and NIH-3T3 cell lines. Interestingly, concentration of AgNPs@Pol (<4 µg/ml) that showed an effective antibacterial effect in combination with methicillin demonstrated low cytotoxicity in cell lines. Therefore, this study provides the optimum concentration range of AgNPs@Pol to treat MRSA, which is < 4 µg/ml.

5. Conclusions

This study demonstrated the synthesis and characterization of AgNPs stabilized with non-ionic, biocompatible polymer, poloxamer exhibiting excellent aqueous stability over three months. The antibacterial activity of AgNPs@Pol against MRSA and MSSA has revealed its significant efficacy against both strains, with a synergistic effect observed in combination with methicillin. The significant increase in ROS production in both bacterial strains after exposure to AgNPs@Pol alone and in combination implies that elevated ROS levels are likely the cause for the antibacterial activity of AgNPs@Pol in *S. aureus*.

The qPCR analysis revealed a notable decrease in gene expression contributing to AMR in MRSA, including *mecA*, *mecR1*, and *femA*. Similarly, the gene expression related to the cell-to-cell adhesion is highly affected by AgNPs@Pol alone or in combination treatment in MRSA. The anti-MRSA effective concentrations of AgNPs@Pol demonstrated low cytotoxicity towards MCF-7, MG-63, and NIH-3T3 cells. Since the AgNPs@Pol are effective against mitotically active cancer cell lines, further studies on primary cells may provide its suitability for clinical applications.

However, the study is limited to short-term *in vitro* testing on a single bacterial species. To comprehensively assess the long-term effectiveness, resistance development, and safety of AgNPs@Pol, further research in animal models and clinical studies is essential. To begin with, 3D cell culture models with infection study may provide more physiologically relevant insights into the treatment of MRSA using AgNPs@Pol. Additionally, using *Drosophila*, *Zebrafish*, and rat models for efficacy assessment and pharmacokinetic studies could facilitate the integration of AgNPs@Pol into existing treatment paradigms. By harnessing the unique properties of AgNPs@Pol, we can pave the way for innovative therapeutic strategies against bacterial infections, addressing the urgent need for novel antimicrobial agents in the face of emerging multidrug-resistant pathogens.

CRedit authorship contribution statement

Nijil S: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Sinchana G Bhat:** Validation, Methodology, Investigation, Formal analysis, Data curation. **Anushree Kedla:** Methodology, Investigation, Formal analysis, Data curation. **Mahima Rachel Thomas:** Validation, Methodology, Investigation, Formal analysis, Data curation. **Sudarshan Kini:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.micpath.2024.107087>.

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